IEOR 4404 Simulation **Fall 2025**

Homework 8

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Problem 1 Messages arrive at a communications facility according to a Poisson process with rate 2 per hour.

(a) Write down the pseudo-codes for simulating the Poisson process until the 10-th hour, using both the interarrival time generation approach and the conditional representation approach.

(b) Suppose the facility consists of three channels, and an arriving message will either go to a free channel if any of them are free, in which case the corresponding channel becomes busy, or else will be lost if all channels are busy. The amount of time that a message occupies a channel is an exponential random variable with mean 1 hour, independent from everything else. Suppose that the system is initially empty at time 0. By using any of the simulators in part (a) and in addition tracing the message occupancy, use simulation to estimate the probability that all three channels are busy at the 10-th hour. Use a replication size (i.e., number of simulated trajectories) 100.

Problem 2 Suppose customers arrive according to a time inhomogeneous Poisson process with intensity function $\lambda(t) = 2 + \cos(2\pi t)$.

(a) Write down the pseudo-codes for simulating the process until time 10, using the interarrival time generation approach, the conditional representation approach, and the modified conditional representation approach that does not require acceptance-rejection.

(b) Implement the three approaches in part (a) to estimate the expected time-averaged number of customers during the time $[0, 10]$, i.e., the expectation of $\frac{1}{10}\int_0^{10} N(t)dt$. For each approach, use a replication size 100.

Problem 3 Consider a $G/G/1$ queue. More specifically, suppose the interarrival times are i.i.d. $Exp(1/2)$ random variables, service times are i.i.d. $Gamma(3, 2)$ random variables with density function given by

$$f(x) = 4x^2e^{-2x}, \quad x \geq 0,$$

and the interarrival and service times are all independent. The server serves customers in a first-come-first-serve order. Write the pseudo-code for simulating the long-run average waiting time of the customers. Implement the code in the computer by simulating 1000 customers, assuming the system is empty initially, and using a burn-in of 100 customers. You may use the fact that $Gamma(3, 2)$ can be represented as the sum of 3 i.i.d $Exp(2)$ random variables.

Problem 4 Consider simulating $P(X > 10)$ where $X \sim Gamma(2, 1)$. You are given that $Gamma(\alpha, \beta)$ has a density

$$\frac{\beta^\alpha}{\Gamma(\alpha)}x^{\alpha-1}e^{-\beta x}, \quad x > 0$$

where Γ is the gamma function. The mean of $Gamma(\alpha, \beta)$ is α/β and the moment generating function $Ee^{\theta X}$ is $(1 - \theta/\beta)^{-\alpha}$ defined for $\theta < \beta$.

(a) Derive the output of one replication obtained from importance sampling by tuning the α parameter so that the mean of the gamma distribution matches exactly 10.

(b) Derive the output of one replication obtained from importance sampling by an exponential tilting of $Gamma(2, 1)$ so that its mean matches exactly 10.

(c) Consider simply using a shifted gamma distribution to speed up the simulation, i.e., use the random variable $X+10$, where $X \sim Gamma(2, 1)$, as the importance sampling variable. Is this a legitimate scheme? Briefly explain your answer.

Problem 5 Consider a single-server queue, where arrivals follow a Poisson process with rate 2 per minute and service times are exponentially distributed with mean 1 minute. Let T_i denote the amount of time that customer i spends in the system. We are interested in using simulation to estimate $E[T_1 + \ldots + T_{10}]$.

- Run a naive Monte Carlo simulation with 100 replications. Give a point estimate of the target quantity and an estimate of the variance per replication.
- Run a simulation with 100 replications using the control variate $\sum_{i=1}^{10} S_i$, where S_i is the ith service time. Give a point estimate of the target quantity and an estimate of the variance per replication, and describe the estimation improvement over the naive estimate in part (a).
- Run a simulation with 100 replications using the control variate $\sum_{i=1}^{10} S_i - \sum_{i=1}^{10} X_{i+1}$, where X_{i+1} is the interarrival time between the ith and $(i+1)$th arrivals. Give a point estimate of the target quantity and an estimate of the variance per replication, and describe the estimation quality compared to the naive estimate in part (a) and the control variate estimate in part (b).

Problem 6 A bank has a portfolio of $N = 100$ loans to N companies and want to evaluate its credit risk. Given that company n defaults, the loss for the bank X_n follows $N(3, 1)$. Defaults are described by indicator variables $D_1, D_2, \ldots, D_N$, with a random background default probability P so that given $P = p$, D_n are i.i.d. Bernoulli random variables with parameter p. P itself follows a $Beta(1, 19)$ distribution with density function given by $f(p) = 19(1-p)^{18}, 0 < p < 1$. Estimate $P(L > x)$, where $L = \sum_{n=1}^{N} D_n X_n$ is the total loss, and $x = 3E[L] = 3NE[P]E[X_n] = 3 \cdot 100 \cdot 0.05 \cdot 3 = 45$. Run naive Monte Carlo simulation and conditional Monte Carlo. In each approach, use 100 replications, and give a point estimate of the target quantity and an estimate of the variance per replication. Comment briefly on their performance comparisons.